

THE SCOTS COLLEGE



YEAR 12

EXTENSION 2 MATHEMATICS

ASSESSMENT TASK 1

FEBRUARY 2004

WEIGHTING: 10%

TIME ALLOWED: 50 MINUTES

INSTRUCTIONS:

- ALL NECESSARY WORKING MUST BE SHOWN.
- BOARD APPROVED CALCULATORS MAY BE USED.
- START EACH QUESTION ON A NEW PAGE.

OUTCOMES:

- E3 :** Uses the relationship between algebraic and geometric representations of complex numbers and of conics.
- E6 :** Combines the ideas of algebra and calculus to determine the important features of the graphs of a wide variety of functions.

QUESTION 1**[MARKS]**

(a) (i) Express $-1 + \sqrt{3}i$ in modulus - argument form. **[1]**

(ii) Hence evaluate $(-1 + \sqrt{3}i)^9$ **[2]**

(b) (i) Find the value of the product $(-1 + \sqrt{3}i)(1 + i)$ **[1]**

(ii) Hence, or otherwise, find the exact value of $\cos \frac{11\pi}{12}$. **[2]**

(c) Sketch on separate Argand diagrams, the locus of z defined by:

(i) $\arg\left(\frac{z-i}{z+1}\right) = 0$ **[2]**

(ii) $|z - (2 + 3i)| = 5$ **[2]**

(d) If $z_1 = a + ib$ and $z_2 = c + id$

(i) Show algebraically that $|z_1 z_2| = |z_1| |z_2|$ and from the Argand diagram explain why $|z_1 + z_2| \leq |z_1| + |z_2|$. **[4]**

(ii) Hence or otherwise, if $|z| \leq \frac{1}{2}$, show that $|(1 + i)z^3 + iz| < \frac{3}{4}$. **[3]**

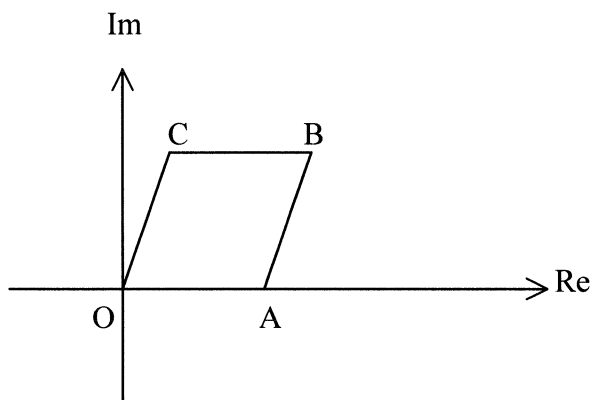
(e) $1 - 2i$ is the root of the equation $z^2 - (3 + i)z + c = 0$.

(i) Explain why the conjugate $1 + 2i$ cannot be a root to the equation. [1]

(ii) Show that the other root is $2 + 3i$. [1]

(iii) Find the value of c . [1]

(f) OABC is a rhombus.
O lies at the origin.
A is on the real axis.
C corresponds to the complex number $1 + \sqrt{3}i$.



(i) Find the complex number corresponding to the point B. [1]

(ii) If the figure is rotated anti-clockwise by $\frac{\pi}{3}$ radians about O to form a new rhombus OA'B'C', draw this new rhombus on an Argand diagram and find the complex number corresponding to B'. [2]

(a) Let $f(x) = (x-1)(x-3)^2$. Sketch each of the following on separate diagrams:

(i) $y = f(x)$ [1]

(ii) $|y| = f(x)$ [3]

(iii) $y = f(|x|)$ [2]

(iv) $y^2 = f(x)$ [3]

(b) Consider the curve $f(x) = 4x^2(2-x^2)$

(i) Find the stationary points of $y = f(x)$ and determine their nature. [3]

(ii) Sketch the graph of $y = f(x)$, showing the x intercepts. [2]

(iii) Also sketch $y = \ln f(x)$ [3]

Ext 2 Maths.
SOLUTIONS - FEBRUARY 2004

QUESTION 1

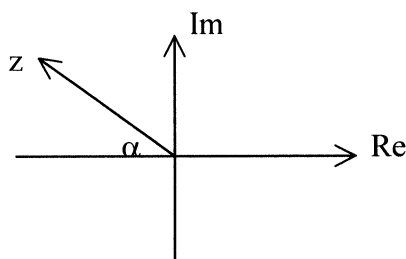
(a) (i) $2 = -1 + \sqrt{3}i$

$$|z| = \sqrt{1+3}$$

$$= 2$$

$$\tan \alpha = \frac{\sqrt{3}}{1}$$

$$\alpha = \frac{\pi}{3}$$



$$\arg(z) = \pi - \frac{\pi}{3}$$

$$= \frac{2\pi}{3}$$

$$\therefore z = 2\text{cis} \frac{2\pi}{3}$$

[MARKS]

½ for mod

½ for arg

(ii) $(-1 + \sqrt{3}i)^9 = 2^9 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right)^9$

$$= 2^9 (\cos 6\pi + i \sin 6\pi)$$

$$= 2^9 = 512$$

1 De Moivre's
1 answer

(b) (i) $(-1 + \sqrt{3}i)(1 + i)$

$$= -1 - i + \sqrt{3}i - \sqrt{3}$$

$$= (-1 - \sqrt{3}) + (\sqrt{3} - 1)i$$

½
½ answer

(ii) $(-1 + \sqrt{3}i)(1 + i)$

$$= 2 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right) \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$$

$$= 2\sqrt{2} \left(\cos \left(\frac{2\pi}{3} + \frac{\pi}{4} \right) + i \sin \left(\frac{2\pi}{3} + \frac{\pi}{4} \right) \right)$$

$$= 2\sqrt{2} \left(\cos \frac{11\pi}{12} + i \sin \frac{11\pi}{12} \right)$$

½

½

equating real parts

$$2\sqrt{2} \cos \frac{11\pi}{12} = -1 - \sqrt{3}$$

½

$$\therefore \cos \frac{11\pi}{12} = \frac{-1 - \sqrt{3}}{2\sqrt{2}}$$

½

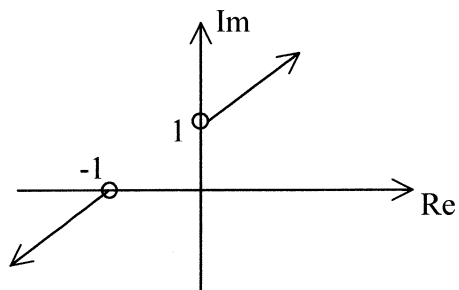
$$= \frac{-\sqrt{2} - \sqrt{6}}{4}$$

(c) (i) $\arg\left(\frac{z-i}{z+1}\right) = 0$

 $\frac{1}{2}$

$$\arg(z-i) - \arg(z+1) = 0$$

i.e. $\arg(z-i) = \arg(z+1)$

 $\frac{1}{2}$ 

1

$\therefore$ line $y = x + 1$ but not in the interval $-1 < x < 0$.

(ii) $|z - (2 + 3i)| = 25$

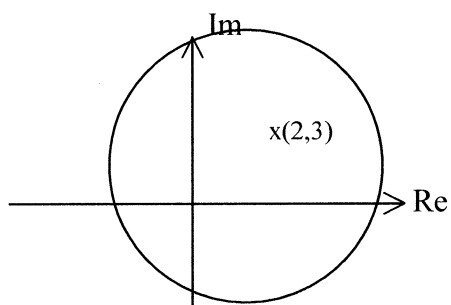
 $\frac{1}{2}$

$$|(x-2) + (y-3)i| = 25$$

$$(x-2)^2 + (y-3)^2 = 25$$

 $\frac{1}{2}$

$\therefore$ circle centre (2, 3) radius 5 units

 $\frac{1}{2}$  $\frac{1}{2}$

(d) (i) $z_1 = a + ib, z_2 = c + id$

$$\text{LHS} = |z_1 z_2|$$

$$= |(a + ib)(c + id)|$$

1

$$= |ac + adi + bci - bd|$$

$$= |(ac - bd) + i(d + bc)|$$

$$= \sqrt{a^2 c^2 - 2abcd + b^2 d^2 + a^2 d^2 + 2abcd + b^2 c^2}$$

1

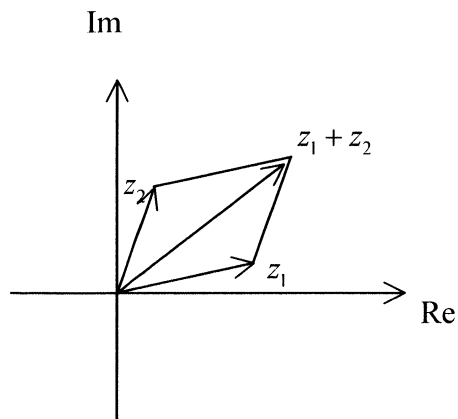
$$= \sqrt{a^2(c^2 + d^2) + b^2(c^2 + d^2)}$$

$$= \sqrt{(a^2 + b^2)(c^2 + d^2)}$$

1

$$= \sqrt{a^2 + b^2} \cdot \sqrt{c^2 + d^2}$$

$$= |z_1| \cdot |z_2|$$

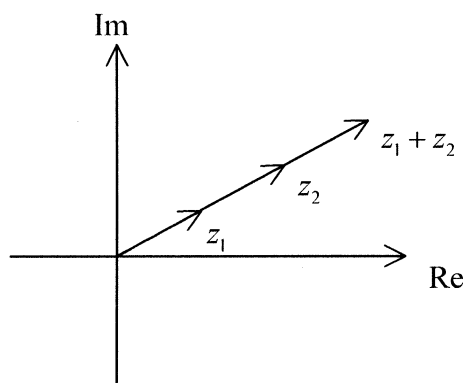


The longest side of a Δ is less than the sum of the other 2 sides.

$$\text{i.e. } |z_1 + z_2| < |z_1| + |z_2|$$

[MARKS]

$\frac{1}{2}$



However, when the $\arg(z_1) = \arg(z_2)$ then

$$|z_1 + z_2| = |z_1| + |z_2|$$

$$|z_1 + z_2| \leq |z_1| + |z_2|$$

$\frac{1}{2}$

$$\begin{aligned} \text{(ii)} \quad & |(1+i)z^3 + iz| \\ &= |z| |z^2(1+i) + i| \end{aligned}$$

$\frac{1}{2}$

$$\leq \frac{1}{2} [|z^2| |1+i| + |i|]$$

$\frac{1}{2}$

$$\leq \frac{1}{2} \left[\frac{1}{4} (\sqrt{2} + 1) \right]$$

$\frac{1}{2}$

$$\leq \frac{1}{2} \left(\frac{\sqrt{2} + 4}{4} \right)$$

$\frac{1}{2}$

$$= \frac{\sqrt{2} + 4}{8}$$

$$< \frac{6}{8} \quad \text{since } \sqrt{2} = 1.42 \dots\dots$$

$\frac{1}{2}$

$$< \frac{3}{4}$$

$\frac{1}{2}$

$$\text{(e)} \quad z^2 - (3+i)z + C = 0$$

(i) Conjugate $1+2i$ cannot be a root since the co-efficient are not real.

1

(ii) Let roots be $\alpha, (1-2i)$

$$\text{sum of roots} = \frac{-b}{a}$$

$\frac{1}{2}$

$$\alpha + 1 - 2i = 3 + i$$

$\frac{1}{2}$

$$\alpha = 2 + 3i$$

(iii) Product of roots $= \frac{c}{a}$

$$(1-2i)(2+3i) = c$$

$$2+3i-4i+6 = c$$

$$\therefore c = 8-i$$

 $\frac{1}{2}$ $\frac{1}{2}$

(f) (i) Let B be $z = x + iy$

$$|z| = \sqrt{1^2 + \sqrt{3}^2} \quad \arg(z) = \frac{\pi}{3}$$

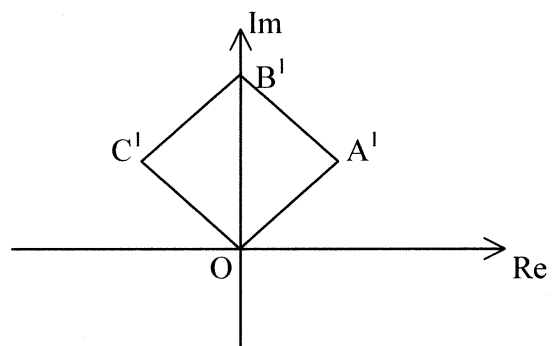
$$= 2$$

 $\frac{1}{2}$

$$\therefore B \text{ is point } 1 + \sqrt{3}i + 2 = 3 + \sqrt{3}i$$

 $\frac{1}{2}$

(ii)



1 diagram

$$\begin{aligned} OB &= \sqrt{3^2 + (\sqrt{3})^2} \\ &= \sqrt{12} \\ &= 2\sqrt{3} \end{aligned}$$

 $\frac{1}{2}$

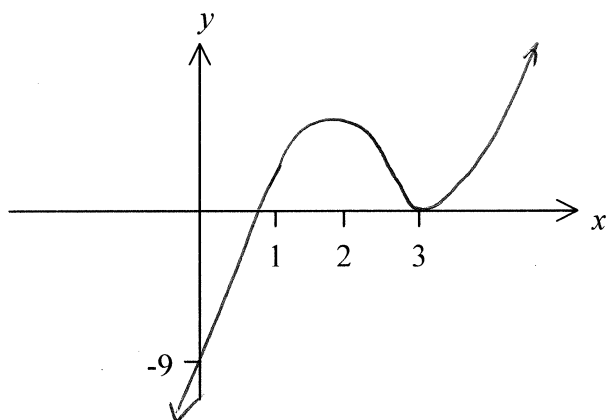
$$\therefore B' \text{ is } 2\sqrt{3}i$$

 $\frac{1}{2}$

QUESTION 2

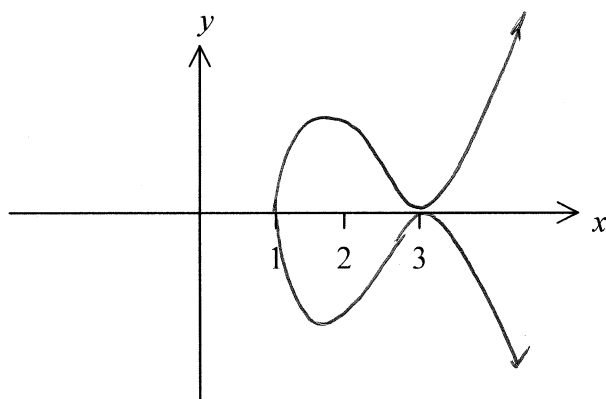
[MARKS]

(a) (i) $y = f(x) = (x-1)(x-3)^2$



1
diagram

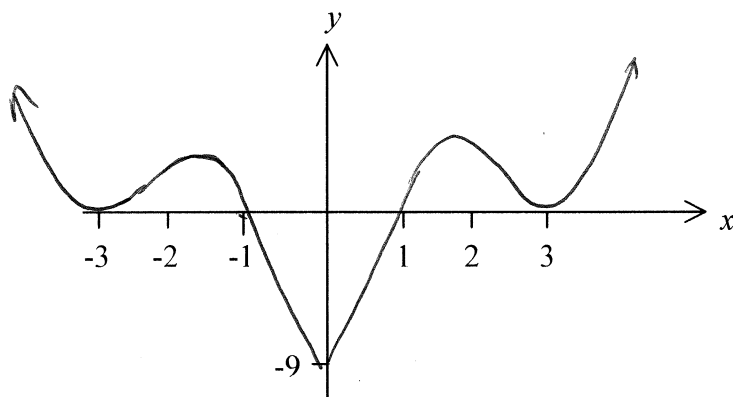
(ii) $|y| = f(x)$



3
correct
diagram

2
half graph

(iii) $y = f(|x|)$



2
correct
diagram

1
variations

[MARKS]

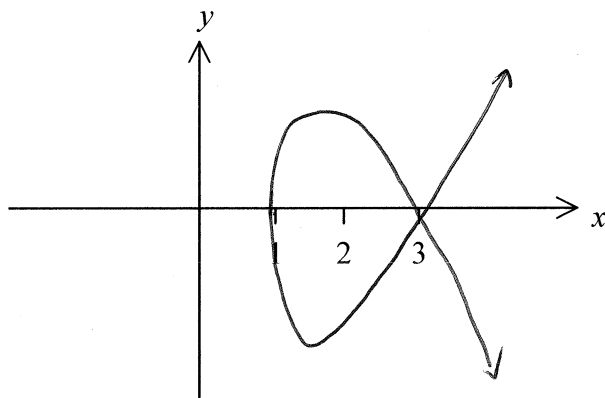
(iv) $y^2 = f(x)$

$$2y \frac{dy}{dx} = f'(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{2y}$$

$\therefore$ Stat pts at $f'(x) = 0, x = 2$

$\therefore$ critical pts, $y = 0, x = 1, 3$



3
correct
diagram

[MARKS]

(b) $f(x) = 4x^2(2 - x^2) = 8x^2 - 4x^4$

(i) $f'(x) = 16x - 16x^3$

$f''(x) = 16 - 48x^2$

For stationary points, $f'(x) = 0$

$16x(1 - x^2) = 0$

$x = 0, x = 1, x = -1$

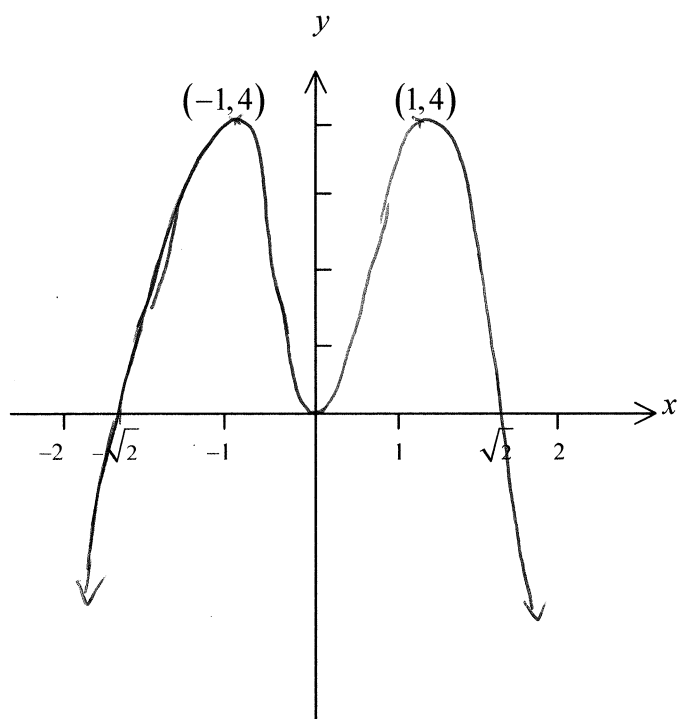
At $x = -1, f(x) = 4, f''(x) < 0 \cap$
 $(-1, 4)$ max turning point.

At $x = 0, f(x) = 0, f''(x) > 0 \cup$
 $(0, 0)$ min turning point.

At $x = 1, f(x) = 4, f''(x) < 0 \cap$
 $(1, 4)$ max turning point.

(ii) Sub $y = 0, 4x^2(2 - x^2) = 0$

$x = 0, x = \pm\sqrt{2}$



1

1st + 2nd
derivatives

1

find stat points

1

determining
their nature

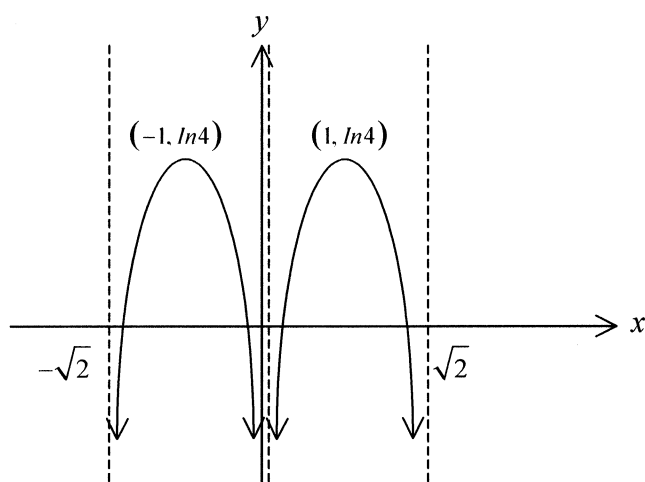
2

complete graph
+ intercepts

[MARKS]

(iii) $y = \ln f(x)$

Domain $-\sqrt{2} < x < 0$ and $0 < x < \sqrt{2}$



1
asymptotes

1
shape of graph

1
scale

$$\frac{dy}{dx} = \frac{f'(x)}{f(x)}$$

Stat points,

$$f'(x) = 0,$$

i.e. $x \neq 0$,

$$x = \pm 1$$

Critical points

$$f(x=0),$$

i.e. $x = 0$,

$$x = \pm\sqrt{2}$$